Arithmetic Function Analogs for K-Tuples

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Abstract

We encode the essential information of k-tuples in some extensions of common arithmetic functions. Analogs of some fundamental relationships between those common functions are shown to hold for our extensions. The basic analytic properties of some of these functions are derived and related to the first Hardy-Littlewood conjecture.

1 K-Tuples

1.1 Basic Properties

Definition 1.1. *Define $\mathbb{K}$, the set of **k-tuples**, by*

$$\mathbb{K} := \mathscr{P}(\mathbb{Z}).$$

*We say a k-tuple is **singular** if it contains exactly one element.*

Some authors forbid infinite k-tuples. We denote the number of elements in a set S by $\#S$. The motivation for much of this paper is to extend a definition or result pertaining to integers to the k-tuples. That is, we view traditional number theoretic concepts involving an integer k as being about the singular k-tuple $\{0\}$, and attempt to extend that result in a natural way to nonsingular k-tuples. We also refer to such singular definitions and results collectively as "singular number theory".

Definition 1.2. *For $A, B \in \mathbb{K}$, we define*

$$A + B := \{a + b : a \in A, b \in B\}.$$

For $k \in \mathbb{Z}$ and $A \in \mathbb{K}$, define for convenience $k + A := \{k\} + A$. That is, $k + A$ is the translation of A by k.

We are often concerned with congruence between elements of k-tuples, but the usual definition of congruence is insufficient, as it describes only

exact incidence between numbers relative to a modulus. We are instead concerned with a variety of types of incidence between translated k-tuples and the integers relatively prime to some modulus. The following definition handles the "modulus" aspect of that need.

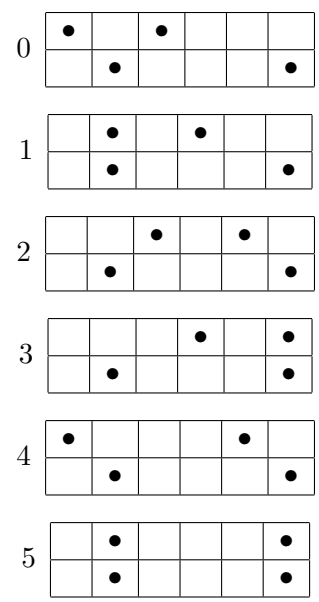
Definition 1.3. *Given a k-tuple A and a natural number n, define*

$$A \bmod n \text{ or } A_n := \{a \bmod n : a \in A\}$$

Define $\mathbb{K}_n$, the set of ***k-tuples modulo n****, to be the set of all such objects.*

1.2 The Totient Function

We now extend Euler's totient function φ to k-tuples Instead of counting which natural numbers up to n are relatively prime to n, we count the translates of a k-tuple A by $k \in \mathbb{Z}_n$ for which all elements are coprime to n. We may also consider counting many types of occurences, and will do so in a later section. We graph the counting process below for $A = \{0, 2\}$ and $n = 6$. The top grid represents a translate of A, and the bottom are those integers up to 6 coprime to 6. Recalling that those coprime integers are periodic with period 6, we simply wrap any elements of the translated A which fall off the end back to 0.



There is only a single occurrence, on table 5, of complete containment, so our totient should return 1 for this A and n.

Definition 1.4 (K-tuple totient)**.**
For a k-tuple A, define φ_A by:

$$\varphi_A(1) := 1$$
$$\varphi_A(n) := \#\{k \in \mathbb{Z}_n : \forall\ a \in A,\ (k+a, n) = 1\}$$

Theorem 1.5. *For any k-tuple A, φ_A is multiplicative.*

Proof. Throughout, let k_n be the class represented by k in $\mathbb{Z}_n$ for any n. Let A be a k-tuple, and let m and n be coprime natural numbers. For each natural number b, define the set $P(b)$ by

$$P(b) = \{k_b \in \mathbb{Z}_b : \forall\ a \in A,\ (k_b + a, b) = 1\}.$$

Then by definition $\#P(a) = \varphi_A(a)$.

Now let $\Gamma : P(mn) \longmapsto P(m) \times P(n)$ be given by

$$\Gamma(k_{mn}) = (k_m, k_n).$$

We will show each of the following:

1. Γ **is well defined.** Let $k_{mn} \in P(mn)$. Then $\forall\ a \in A,\ (k+a, mn) = 1$, so $\forall\ a \in A,\ (k+a, m) = 1$. Proceeding likewise for n, $(k_m, k_n) \in P(m) \times P(n)$, so Γ is well defined.

2. Γ **is injective.** Let $a_{mn}, b_{mn} \in P(mn)$ such that $\Gamma(a_{mn}) = \Gamma(b_{mn})$. Then $a_m = b_m$ and $a_n = b_n$. Because m and n are coprime, $a_{mn} = b_{mn}$. Thus, Γ is injective.

3. Γ **is surjective.** Let $(b_m, c_n) \in P(m) \times P(n)$. By the Chinese Remainder Theorem, there exists $d_{mn} \in \mathbb{Z}_{mn}$ such that $d_m = b_m$ and $d_n = c_n$. It then suffices to show $d_{mn} \in P(mn)$, for this gives $\Gamma(d_{mn}) = (b_m, c_n)$.

   Let $a \in A$. We have

   $$\begin{aligned} & b_m \in P(m) \\ \implies & (a + b_m, m) = 1 \\ \implies & (a + d_m, m) = 1 \\ \implies & (a + d_{mn}, m) = 1. \end{aligned}$$

   Proceeding likewise for c_n, we have $(a + d_{mn}, n) = 1$. Combining these gives $(a + d_{mn}, mn) = 1$, so $d_{mn} \in P(mn)$ and Γ is surjective.

Thus, Γ is a bijection, so

$$\begin{aligned}\varphi_A(mn) &= \#P(mn) \\ &= \#(P(m) \times P(n)) \\ &= \#P(m) \cdot \#P(n) \\ &= \varphi_A(m) \cdot \varphi_A(n).\end{aligned}$$

□

1.3 The Unit Function

The singular unit function $u(n)$ is defined by $u(n) = 1$ for all n. It is surprising that such a trivial function has a non-trivial extension to k-tuples.

Definition 1.6. *For any k-tuple A and prime p, define $u_A(p)$ by:*

$$u_A(p) = \#A_p$$

We extend u_A to all natural numbers by complete multiplicativity.

We have immediately from complete multiplicativity that $u_A(1) = 1$. The definition for primes tells us that u_A gives the number of residues the elements of A fill modulo p. The definition of u_A on primes is used in the first Hardy and Littlewood conjecture, discussed later, and so is not entirely new. It's usage in the following theorem to describe extensions of φ is also known, at least in some well known cases.

Theorem 1.7. *For any k-tuple A, prime p, and natural number k we have:*

$$\varphi_A(p^k) = p^k - u_A(p)p^{k-1}.$$

Proof. By definition we have

$$\begin{aligned}\varphi_A(p^k) &= \#\{j \in \mathbb{Z}_{p^k} : \forall\ a \in A,\ (j + a, p^k) = 1\} \\ &= \#\{j \in \mathbb{Z}_{p^k} : \forall\ a \in A_{p^k},\ (j + a, p^k) = 1\}.\end{aligned}$$

If this condition is satisfied by some j, it will be satisfied by $j + pm$ for any m, and there are p^{k-1} multiples of p in $\mathbb{Z}_{p^k}$. In other words, as A_{p^k} is translated through $\mathbb{Z}_{p^k}$, we gain the same amount towards our total count

as we pass through each period of size p, and there are p^{k-1} such periods. We then have

$$\begin{aligned}\varphi_A(p^k) &= p^{k-1}\#\{j \in \mathbb{Z}_p : \forall\ a \in A_p,\ (j+a,p) = 1\} \\ &= p^{k-1}(p - \#\{j \in \mathbb{Z}_p : \exists\ a \in A_p,\ (j+a,p) \neq 1\})\end{aligned}$$

Now, $(j+a,p) \neq 1$ exactly once per $a \in A_p$ as j ranges over $\mathbb{Z}_p$ (when $j + a \equiv 0 \bmod p$). Thus, the size of the set in the previous equation is $\#A_p$, which gives the required formula. □

Theorem 1.8 (Totient Product Formula)**.**
For any k-tuple A and natural number n, we have:

$$\varphi_A(n) = n \prod_{p|n} \left(1 - \frac{u_A(p)}{p}\right).$$

Proof. Let A be a k-tuple and n be a natural number. Then n has a factorization, say

$$n = \prod_{p_i|n} p_i^{a_i}.$$

Then we have by theorems 1.5 and 1.7,

$$\begin{aligned}\varphi_A(n) &= \varphi_A\left(\prod_{p_i|n} p_i^{a_i}\right) \\ &= \prod_{p_i|n} \varphi_A(p_i^{a_i}) \\ &= \prod_{p_i|n} \left(p_i^{a_i} - u_A(p_i)p_i^{a_i-1}\right) \\ &= n \prod_{p_i|n} \left(1 - \frac{u_A(p_i)}{p_i}\right).\end{aligned}$$

□

Theorem 1.9. *For any k-tuple A, the totient function has the following properties:*

(a) $\varphi_A(mn) = \varphi_A(m)\varphi_A(n)(d/\varphi_A(d))$, *where* $d = (m,n)$.

(b) $\varphi_A(mn) = \varphi_A(m)\varphi_A(n)$ *if* $(m, n) = 1$.

(c) $a|b$ *implies* $\varphi_A(a)|\varphi_A(b)$.

Proof. The proof of these follows nearly verbatim the singular case, which is shown in theorem 2.5 of [1].

□

Definition 1.10. *Let v be an arithmetic function such that:*

1. *v is completely multiplicative.*
2. *v is eventually constant, say $v(p) = k$ for $p > N$.*
3. *For all primes p, $v(p) \leq p$ and $v(p) \leq k$.*

The we say v is ***unit looking****. If there exists a k-tuple such that $u_A = v$, we say A generates v, and say v is good. Otherwise, v is bad.*

Unit looking functions, as far as can be checked by simple methods, appear to have been generated by some k-tuple. For $k = 2$, all unit looking functions are good. Proving any individual unit looking function is bad is a tedious affair involving many cases, and so is omitted. A general algorithm determining this property would be useful in later computation, but it seems unlikely such an algorithm is feasable in the general case. As a concrete example, the unit looking function v with $v(2) = 1$, $v(3) = v(103) = 2$, and $v(p) = 3$ for all other primes p is bad, and is the minimal example. Computation shows that bad unit looking functions become far more common as the primes at which they differ from their eventual value grow, and it seems likely that the proportion of good functions goes to zero in that limit. We may, however, easily find k-tuples whose unit matches a given unit looking function up to an arbitrary bound.

Theorem 1.11. *Let p_n be the nth prime number. Let v be a unit looking function. Then for any natural N, there exists a k-tuple A such that $u_A(n) = v(n)$ for $n \leq N$.*

Proof. By the complete multiplicativity of u_A and v, we need only show the equality on primes up to N.

Let k be the eventual value of v on primes. Let $A_1 = \{0\}$. If $k = 1$, we have $u_{A_1} = v$ and are done. Now for each i from 2 to k, let Q_i be the set of primes up to N for which $u_{A_i}(p) < v(p)$. By the Chinese Remainder Theorem there exists a which solves $a \equiv i \pmod{q}$ for all $q \in Q_i$, and $a \equiv 0$ modulo each other prime up to N. Let $A_i = A_{i-1} \cup a$.

Then for each prime p_n up to N, exactly the first $v(p_n)$ residues have been filled by elements of A_k. That is, $u_{A_k}(p_n) = v(p_n)$. □

1.4 Dirichlet Convolution

For any arithmetic functions f and g we denote their Dirichlet product by

$$(f * g)(n) := \sum_{ab=n} f(a)g(b).$$

We denote the Dirichlet inverse of an arithmetic function f by f^{-1}. We define the arithmetic function N by $N(n) = n$.

Definition 1.12. *For any k-tuple A, define μ_A to be the unique Dirichlet inverse of u_A.*

By definition u_A is completely multiplicative, so by Theorem 2.17 in [1], we have its Dirichlet inverse to be

$$\mu_A(n) = u_A^{-1}(n) = \mu(n)u_A(n).$$

We have also from Theorem 2.18 of [1] that for any multiplicative function f and natural number n, that

$$\sum_{d|n} \mu(d)f(d) = \prod_{p|n}(1 - f(p)), \tag{1}$$

which is used to prove the following.

Theorem 1.13. *For any k-tuple A, we have*

$$u_A * \varphi_A = N.$$

Proof. Applying equation (1) with $f(n) = u_A(n)/n$ to the totient product formula, we have

$$\begin{aligned}
\varphi_A(n) &= n \prod_{p|n} \left(1 - \frac{u_A(p)}{p}\right) \\
&= n \sum_{d|n} \mu(d) \frac{u_A(d)}{d} \\
&= \sum_{d|n} \mu_A(d) \frac{n}{d} \\
&= (\mu_A * N)(n).
\end{aligned}$$

Taking the Dirichlet product of both sides by u_A, we obtain the theorem. □

This result, while simple, is the reason for our later analysis. Intuitively, φ_A should encode information about relative primality in the k-tuple sense. The function N is neutral in this regard, so we should expect $u_A = N * \varphi_A^{-1}$ to retain some aspect of that information while playing nicely with Dirichlet series.

1.5 Alternatives to φ_A

When we defined φ_A, we counted occurences of complete coprimality. We could also ask for the occurences of any amount of coprimality, a specific amount, or a minimum amount, among other requirements. However, these examples do not produce multiplicative φ analogs.

There is also a subtlety in defining them that doesn't exist in our φ_A. We defined φ_A in terms of translates of A, but we could have equivalently counted the translates of A_n. These definitions are inequivalent for the examples listed above, for we will have $\#A_n < \#A$ when elements of A coincide modulo n. That is, the issue with those examples is that they rely on the number of $a \in k + A$ or $a \in k + A_n$ to satisfy a property, which may differ. We may avoid this issue be only considering functions which count those k such that each of the elements of $k + A$ possesses some property, rather considering a property on $k + A$ itself. But to consider a function to be an analog of φ, the property considered should be "Are these elements coprime?". There is only one function of this type which considers this property: φ_A.

We may relax our constraints, though, on what property constitutes a totient to depend on the value of $(k + a, n)$, rather than if it is 1 or not. This gives not only additional k-tuple totients, but also additional singular totients. In general, we may use the property that $(k + a, n)$ is limited to a specific subset of the primes and obtain a multiplicative totient.

Definition 1.14. *Given a set of primes B, let $a \prec B$ mean that a is composed only of elements of B.*

In particular, if this set is exactly the primes less than some bound b, we are requiring $(k + a, n)$ to be b-smooth:

Definition 1.15. *If the prime factors of an integer k are bounded by b, we say k is b-**smooth**.*

Definition 1.16. *For a k-tuple A and a set of primes B, define $\varphi_{A,B}$ by*

$$\varphi_{A,B}(1) := 1,$$
$$\varphi_{A,B}(n) := \#\{k \in \mathbb{Z}_n : \forall\ a \in A,\ (k + a, n) \prec B\}.$$

Theorem 1.17. *For any k-tuple A and set of primes B, $\varphi_{A,B}$ is multiplicative.*

Proof. The proof follows exactly that of φ_A and so is omitted. □

What, then, are the corresponding unit functions $u_{A,B}$? They are the same as our earlier u_A, except they are zero at every number containing a factor in B. They are completely multiplicative, can be used to compute $\varphi_{A,B}$ in the same way, and yield $u_{A,B} * \varphi_{A,B} = N$. The proofs of these facts follow exactly those for u_A. We return now to the realm of φ_A, keeping in mind that many results may be extended trivially in this way to $\varphi_{A,B}$.

The earlier nonmultiplicative examples are not without value, and not beyond hope. In essentially all cases, we can express the desired function as a linear combination of φ_A. Consider first for some n and k-tuple A, counting the $k \in \mathbb{Z}_n$ such that exactly q elements of $k + A$ are coprime to n. We call the value q the quota, and refer to this as the unwrapped, exact quota case.

Definition 1.18 (Quota k-tuple totients)**.** *For any k-tuple A and quota $q \in \mathbb{Z}$ with $0 \leq q \leq \#A$, define $\Upsilon_A^{q=}$ by*

$$\begin{aligned}
&\Upsilon_A^{q=}(1) := 1 \\
&\Upsilon_A^{q=}(n) := \#\{k \in \mathbb{Z}_n : q = \#\{a \in A : (k + a, n) = 1\}\}.
\end{aligned}$$

Replacing A with A_n in this definition yields a wrapped case, and requiring at least q values to be coprime to n yields a minimum quota case. We denote the wrapped with ϱ rather than Υ, and minimum quotas use $q \leq$ in the superscript.

Theorem 1.19. *For any k-tuple A and quota q,*

$$\Upsilon_A^{q=} = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{q}{k} \sum_{\substack{B \subset A \\ \#B = k}} \varphi_B$$

Proof. TODO □

Theorem 1.20. *For any k-tuple A, and quota q,*

$$\Upsilon_A^{q\leq} = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{k-1}{k-q} \sum_{\substack{B \subset A \\ \#B = k}} \varphi_B.$$

Proof. TODO $\square$

The formulae for the corresponding ϱ functions are identical: simply replace A with A_n throughout. This creates a dependance on n in the final sum which appears to explode the computational complexity, but for sufficiently large n, $A = A_n$, so the order of computational complexity is the same. This also shows that the wrapped and unwrapped totients are eventually equivalent, as should be expected. We will make no further use of the wrapped totients, and the proofs are mostly the same as the unwrapped cases, so they are omitted.

1.6 Totients as Convolutions

We may also obtain many totient-like functions in terms of discrete modular convolutions.

Definition 1.21. *Given $f, g : \mathbb{Z}_n \to \mathbb{C}$, define the discrete modular convolution $\odot_n$ by*

$$(f \odot g)(k) = \sum_{\substack{a+b \equiv k \\ \bmod n}} f(a)g(b),$$

where $a, b \in \mathbb{Z}_n$.

The symbol $\odot$ is non-standard. Normally $*$ would be used, but it is already in use for Dirichlet convolution.

Research line

We get the totients by asking questions about the convolution. For φ_A, the question is "how many times did we get to $\#A_n$?". Everything else is a question about how many times different values were obtained. The questions are so similar to the original definitions, it seems like the convolution is just a way to describe the queried set as a function. That is, instead of asking for the number of k for which $q = \#S$, we are asking for the number of k for which a convolution takes on certain value q times.

Another convolution centric idea, we have $\varphi_A(n) = \#(\varkappa_n +_n A)$

1.7 Von Mangoldt Extensions

We could have indroduced u_A or φ_A from the other via the formula $u_A * \varphi_A = N$, though we would then need to prove the definitional properties of whichever function came second. We take that direction now, extending the Von Mangoldt function Λ via a formula, and then finding what properties it possesses.

Definition 1.22. *Let A be a k-tuple. Define Λ_A by*

$$\Lambda_A := \log * \mu_A.$$

TODO: And then? Below this line in this section is temporary while researching. These functions often behave like the base version. On primes they are the same. On powers of primes, you can only obtain logs of powers of the prime (obv), but it seems like it is limited in how many of those powers are available? e.g., at 9 we are getting only 1,0 powers, at 25 we get 1,0,-1,-2 powers. The default behavior is 1 power. Thinking for one second, yeah no doy, the power is always $u_A(p) + k(1 - u_A(p))$. I'm not sold on this having any meaning outside its definition. I think the definition that leads to HLC is

$$\Lambda_A(n) = \prod_{k \in n+A} \Lambda(k).$$

Maybe there is some asymptotic link between this product version and a non-product weighted version? We can describe this as a pointwise product of shifted copies of Λ. Seems to me that this whole mess (number theory) has something to do with the interplay between the pointwise product, (general) convolution products, and addition over sequences. The zeta function (algebraicly) is the pointwise product identity put into the product convolution. The product convolution identity, put into the pointwise product, just returns the product convolution identity times $f(1)$, which is somewhat less interesting than the zeta function.

AFAIK only the sum and product convolutions have generating functions, though I should look into proving that, or considering other ways of forcing analytic data our of general conv. If that's possible, those generating functions at u may be as interesting as zeta or geometric series. Anyway this belongs in the convolution paper.

1.8 Admissibility

Definition 1.23. *We say a k-tuple A is* ***admissible*** *if, for each prime p, we have*

$$u_A(p) < p.$$

Then a k-tuple is admissible if it does not fill all of the residue classes modulo any prime. We are now equipped to state the prime k-tuple conjecture.

Conjecture 1.24 (Prime K-Tuple Conjecture)**.** *For any finite, admissible k-tuple A, there exists an infinite k-tuple B such that $A + B \subset \mathbb{P}$.*

Admissibility is required, for if a k-tuple A was inadmissible by prime p, for $b \in B$ large enough that each element of $b+A$ is greater than p, we must have some element of A filling the 0 residue. That element is then divisible by p, $b+A \not\subset \mathbb{P}$, and it cannot be for any b beyond that bound. Thus, there can be only finitely many integers at which $b+A \subset \mathbb{P}$ and no valid B exists.

The finiteness of A is also required. For example, let A be the set of all primorials. This fills at most i residues of each prime p_i and so is admissible. We cannot have negatives in B, for $2 \in A$ is the minimum prime. We cannot have 0 or 1 in B, because $A+B$ would contain 6 and 4 respectively. If B contains $b > 1$, b is divisible by some prime p, and the primorials will eventually be divisible by p. Then for some element $a \in A$ divisible by p, $p|a+b$, so $A+B \not\subset \mathbb{P}$. Thus, there are no integers b such that $b+A \subset \mathbb{P}$. Not only can B not be infinite in this case, it cannot contain any integer.

If the conjecture holds, there exist infinite k-tuples A with arbitrary large finite B such that $A+B \subset \mathbb{P}$. Indeed, given a finite admissible k-tuple B, we apply the conjecture and find such an A. The question of adding two infinite k-tuples and obtaining a prime k-tuple is handled in greater generality in [2], the results of which we state in less generality as the following theorem. It is often said that the Prime K-Tuple Conjecture feels intuitively true, so it is somewhat alarming that it holds as a consequence the following result, which seems intuitively impossible.

Theorem 1.25. *Assume the Prime K-Tuple Conjecture. Then there exist infinite k-tuples A and B such that $A+B \subset \mathbb{P}$.*

2 Analytic Results

We denote the Derichlet series of an arithmetic function f by

$$\mathcal{D}(f)(s) := \sum_{n \geq 1} \frac{f(n)}{n^s}.$$

The function u_A is convenient to work with in the setting of arithmetic functions in that it takes on only integer values and gives the relationship $u_A * \phi_A = N$. Transporting it to the domain of Dirichlet series, though, poses a challenge. The function $\mathcal{D}(u_A)(s) = \sum_{n \geq 1} u_A(n) n^{-s}$ permits an Euler product, and that product can be written as a power of ζ, a finite product, and the series $\mathcal{D}((\#A)^{\Omega})$, where $\Omega(n)$ is the number of primes (counting multiplicity) which compose n. While the final portion of this form has reasonable convergence properties, it will be the cause of much

of the complexity in analyzing u_A. That portion, though, does not seem to have much to do with the additive patterns we seek to analyze; rather it describes that u_A should grow approximately with the number of primes in its composition. The following adjustment, then, gives an arithmetic function which contains the same essential information, but which has far simpler analytic properties.

Definition 2.1. *Let A be a k-tuple. For each n, define $w_A(n)$ by*

$$w_A(n) = \frac{u_A(n)}{(\#A)^{\Omega(n)}}.$$

This is equivalent to defining $w_A(p) = u_A(p)/\#A$ on primes and extending it by complete multiplicativity as with u_A, but we may also show this directly: because Ω is completely additive, $(\#A)^{\Omega}$ is completely multiplicative, so w_A is as well. Note also that for primes p with $\max(A) \leq p$, $u_A(p) = \#A$, so w_A is eventually 1 on primes. We will work with both $\mathcal{D}(u_A)$ and $\mathcal{D}(w_A)$.

2.1 The Dirichlet Series of w_A

We have that w_A is completely multiplicative, so $\mathcal{D}(w_A)$ permits the Euler product

$$\mathcal{D}(w_A)(s) = \prod_p (1 - w_A(p)p^{-s})^{-1},$$

but we know for sufficiently large primes, $w_A(p) = 1$, so letting S_A be the set of primes on which $w_A(p) \neq 1$, we have

$$\mathcal{D}(w_A)(s) = \zeta(s) \prod_{p \in S_A} \frac{1 - p^{-s}}{1 - w_A(p)p^{-s}}.$$

Let $M_A(s)$ denote the finite product above and call this the modulator for A. This form immediately yields the analytic continuation of $\mathcal{D}(w_A)$ via the continuation of ζ to the entire plane, except at the pole contributed by ζ at $s = 1$, and one infinite (but well behaved) set of poles for each $p \in S_A$. These poles are at the solutions of $0 = 1 - w_A(p)p^{-s}$, which are found, for any $k \in \mathbb{Z}$, at $s_{p,k} = \frac{\log(w_A(p)) + 2\pi k i}{\log(p)}$.

Proposition 2.2. *For any admissible k-tuple, we have*

$$\frac{1}{x} \sum_{n \leq x} w_A(n) \sim M_A(1).$$

Proof. Every modulator pole has real part less than one. Thus, we can apply the Wiener-Ikehara theorem (1.1 in [3]) which gives

$$\frac{1}{n}\sum_{n\leq x} w_A(n) \sim \mathrm{Res}(\mathcal{D}(w_A), 1).$$

We have $\mathrm{Res}(\zeta, 1) = 1$, so

$$\mathrm{Res}(\mathcal{D}(w_A), 1) = M_A(1),$$

as was needed. □

To find the residues of any pole of the modulator, we first show that no two poles among those listed coincide. Clearly none of those given by the modulator coincide with $s = 1$, so we need only check that no poles among the modulator coincide. Suprisingly, though, this will walk us directly into the four exponentials conjecture.

Conjecture 2.3 (Four exponentials conjecture)**.** *For $1 \leq i, j \leq 2$, let $\lambda_{ij} \in \mathbb{C}$ such that each $exp(\lambda_{ij})$ is algebraic. Suppose that λ_{11} and λ_{12} are linearly independent, and that λ_{21} and λ_{22} are linearly independent, both in the rationals. Then*

$$\lambda_{11}\lambda_{22} \neq \lambda_{12}\lambda_{21}.$$

Lemma 2.4. *Assume the four exponentials conjecture. Then we have*

$$s_{p_1,k_1} = s_{p_2,k_2} \implies (p_1, k_1) = (p_2, k_2).$$

Proof. Assume $s_{p_1,k_1} = s_{p_2,k_2}$, and assume for contradiction $p_1 \neq p_2$ Then their real parts coincide and we have $\log a_1/\log p_1 = \log a_2/\log p_2$, which gives

$$\log a_1 \log p_2 = \log a_2 \log p_1. \tag{2}$$

Each a_i and p_i is rational and thus algebraic, so the exponential of each logarithm is algebraic.

To show the linear independence of $\log p_1$ and $\log p_2$, assume there are $r, s \in \mathbb{Q}$ such that $r \log p_1 + s \log p_2 = 0$. Then $p_1^r p_2^s = 1$, which is only possible if $r = s = 0$.

If $\log a_1$ and $\log a_2$ are dependent, then there are $r, s \in \mathbb{Q}$, not both 0, such that $r \log a_1 + s \log a_2 = 0$. Without loss of generality, assume $r \neq 0$. Then $a_1 = a_2^{-s/r}$. Applying this to equation (2), we have

$$-\frac{s}{r}\log p_2 = \log p_1$$
$$\implies p_2^{-s/r} = p_1,$$

which is a contradiction. Thus, $\log a_1$ and $\log a_2$ are linearly independent.

Finally, we may apply the four exponentials conjecture, which contradicts (2). Thus, $p_1 = p_2$. Showing $k_1 = k_2$ is trivial. $\square$

The values of a_i in the above proof are not entirely arbitrary, coming from w_A. As such, there may be some way of showing this result holds without the four exponentials conjecture. We continue to assume the conjecture as we now find the residues of the poles associated with the modulator.

Theorem 2.5.

$$Res(\mathcal{D}(w_A), s_{p,k}) = \frac{1 - p^{-s_{p,k}}}{\log p}\zeta(s_{p,k}) \prod_{\substack{q \in S \\ q \neq p}} \frac{1 - q^{-s_{p_k}}}{1 - w_A(q)q^{-s_{p,k}}}.$$

Proof. By the lemma, each pole of $\mathcal{D}(w_A)$ is simple. That is, for any $s_{p,k}$ we have $\mathcal{D}(w_A)(s) = H_{p,k}(s) \cdot g(s)$, where

$$g(s) = \frac{1}{1 - w_A(p)p^{-s}},$$

and

$$H_{p,k}(s) = \zeta(s)(1 - p^{-s}) \prod_{\substack{q \in S \\ q \neq p}} \frac{1 - q^{-s_{p_k}}}{1 - w_A(q)q^{-s_{p,k}}},$$

where $H_{p,k}$ is holomorphic at $s_{p,k}$. Now, note that by the definition of $s_{p,k}$, $1 - w_A(p)p^{-s_{p,k}} = 0$. Using this, we have

$$\begin{aligned} \mathrm{Res}((1 - w_A(p)p^{-s})^{-1}, s_{p,k}) &= \frac{1}{\frac{d}{ds}(1 - w_A(p)p^{-s})|_{s_{p,k}}} \\ &= \frac{1}{w_A(p)\log(p)p^{-s_{p,k}}} \\ &= \frac{1}{\log(p)}, \end{aligned}$$

which gives

$$\mathrm{Res}(\mathcal{D}(w_A), s_{p,k}) = \frac{H_{p,k}(s_{p,k})}{\log(p)}$$

as was needed. $\square$

2.2 The First Hardy-Littlewood Conjecture

Definition 2.6. *For any admissible k-tuple A, define*

$$\pi_A(x) = \#\{n \leq x : n + A \subset \mathbb{P}\}.$$

Definition 2.7. *For $n > 0$, define*

$$\mathrm{Li}_n(x) = \int_2^\infty \frac{\mathrm{dt}}{\log^n(t)}.$$

We call this the nth (offset) logarithmic integral.

Conjecture 2.8 (First Hardy-Littlewood conjecture)**.** *Let A be an admissible k-tuple. Then*

$$\pi_A(x) \sim C_A \mathrm{Li}_{\#A}(x),$$

where

$$C_A = \prod_p \frac{1 - u_A(p)p^{-1}}{(1 - p^{-1})^{\#A}}.$$

This, of course, implies the prime k-tuple conjecture. It also implies that the exact structure of a k-tuple does not determine its asymptotic behavior, but rather that behavior is entirely decided by u_A. For example, $\{0, 2\}$ and $\{0, 4\}$ produce the same u_A. This yields an equivalence relation.

Definition 2.9. *Given k-tuples A and B, we say they are* ***residually equivalent*** *if $u_A = u_B$.*

The following asymptotic expansion is well known:

$$\mathrm{Li}_1(x) \sim \frac{x}{\log(x)} \sum_{k \geq 0} \frac{k!}{\log^k(x)}. \tag{3}$$

The simpler estimate $\frac{x}{\log x}$ given in the prime number theorem is then the truncation of this expansion to its first term. Then we should expect to analagously use the truncation of the asymptotic expansion of Li_n in attempting to gain traction in the first Hardy-Littlewood conjecture.

Proposition 2.10. *TODO: this result needs additional care due to the meaning of asymptotic expansion, though it is correct. We have the asymptotic expansion of Li_n to be*

$$\mathrm{Li}_n(x) \sim \frac{x}{(n-1)! \log x} \sum_{k \geq n-1} \frac{k!}{\log^k(x)}.$$

Proof. First, apply integration by parts to Li_n with $dv = dx/(x\log^n(x))$ and $u = x$ to obtain

$$\mathrm{Li}_n(x) = \frac{-x}{(n-1)\log^{n-1} x} + \frac{1}{n-1}\mathrm{Li}_{n-1}(x) \tag{4}$$

for $n > 1$. We have the result for $n = 1$: it is exactly equation (3). Assume then for induction the result holds for some n. Then applying equation (4) we have

$$\begin{aligned}
\mathrm{Li}_{n+1}(x) &= \frac{-x}{n\log^n x} + \frac{1}{n}\mathrm{Li}_n(x) \\
&\sim \frac{-x}{n\log^n x} + \frac{x}{n!\log x}\sum_{k\geq n-1}\frac{k!}{\log^k x} \\
&= \frac{x}{n!\log x}\left(\frac{-(n-1)!}{\log^{n-1} x} + \sum_{k\geq n-1}\frac{k!}{\log^k x}\right) \\
&= \frac{x}{n!\log x}\sum_{k\geq n}\frac{k!}{\log^k x}.
\end{aligned}$$

□

All this is to say: if the first Hardy-Littlewood conjecture is correct, we should attempt to show, following in the steps of the prime number theorem, that

$$\pi_A(x) \sim \frac{C_A x}{\log^{\#A} x}.$$

2.3 The Dirichlet Series of u_A

For each admissible constellation A, conjecture 2.8 gives an associated constant C_A. This section aims to explore these constants in our context. Just as we describe $\mathcal{D}(w_A)$ in terms of ζ and a finite modulator, we may describe $\mathcal{D}(u_A)$ as

$$\mathcal{D}(u_A)(s) = \zeta^{\#A}(s)\prod_{p\in S_A}\frac{(1-p^{-s})^{\#A}}{1-u_A(p)p^{-s}}. \tag{5}$$

We refer to the final product once again as the modulator, this time denoted $N_A(s)$, and this time it is infinite. The following result essentially states that the product which yields C_A converges, which is well known, but is given for completeness.

Proposition 2.11. *For any admissible k-tuple A, there exists $\epsilon > 0$ such that $N_A(s)$ converges for $\Re(s) > 1 - \epsilon$.*

Proof. Let A be an admissible k-tuple. Then $u_A(p) < p$, so the real parts of the solutions of $1 - u_A(p)p^{-s} = 0$ are all left of 1, and go to 0 as $p \to \infty$. Then there exists $\epsilon > 0$ such that all solutions of $1 - u_A(p)p^{-s} = 0$ lie to the left of $1 - \epsilon$. Fix $s = \sigma + it$ with $\sigma > 1/2$. Let

$$N_{A,r}(s) = \prod_{p>r} \frac{(1 - p^{-s})^{\#A}}{1 - u_A(p)p^{-s}}.$$

We will show the convergence of $N_{A,r}$ at s for a fixed r, from which the result follows, as the finitely many removed factors all converge for $\Re(s) > 1 - \epsilon$. We fix r to be large enough that both $\#Ar^{-\sigma} < 1/2$, and $u_A(p) = \#A$ for $p > r$. We then have

$$\begin{aligned}\log N_{A,r}(s) &= \log \prod_{p>r} \frac{(1 - p^{-s})^{\#A}}{1 - \#Ap^{-s}} \\ &= \sum_{p>r} \left(\#A \log(1 - p^{-s}) - \log(1 - \#Ap^{-s})\right).\end{aligned}$$

Expanding the logs into power series gives

$$\begin{aligned}\log N_{A,r}(s) &= \sum_{p>r} \left(-\#A \sum_{k\geq 1} \frac{p^{-sk}}{k} + \sum_{k\geq 1} \frac{(\#Ap^{-s})^k}{k} \right) \\ &= \sum_{p>r} \sum_{k\geq 2} \left(\frac{(\#A)^k - \#A}{k} p^{-sk} \right).\end{aligned}$$

Note the cancellation of the $k = 1$ term. Now,

$$|\log N_{A,r}(s)| \leq \sum_{p>r} \sum_{k\geq 2} \left(\#Ap^{-\sigma}\right)^k.$$

For each $p > r$, by our bound on r, the k sum is a convergent geometric series. We have also for $p > r$ that $1 - \#Ap^{-\sigma} > 1/2$, so

$$\begin{aligned}|\log N_{A,r}(s)| &= \sum_{p>r} \frac{(\#Ap^{-\sigma})^2}{1 - \#Ap^{-\sigma}} \\ &\leq 2(\#A)^2 \sum_{p>r} p^{-2\sigma} \\ &\leq 2(\#A)^2 \sum_{n>r} n^{-2\sigma}.\end{aligned}$$

We have $\sigma > 1/2$, so this is a convergent p-series. Thus, $\log N_{A,r}$ converges absolutely, and exponentiating gives the result. $\square$

We do not need the full domain of convergence of the prior result; our goal was to justify evaluating $N_A(1)$ and obtaining the following.

Corollary 2.12. *For any admissible k-tuple A, we have*

$$\frac{1}{C_A} = N_A(1),$$

and

$$Res(\zeta^{\#A}, 1) = C_A Res(\mathcal{D}(u_A), 1).$$

Proof. The first equality is immediate from the definitions of C_A and N_A. Taking residues in (5) then yields the second equality. $\square$

The Laurent expansion of ζ is known, so the residues above are computable. We may factor N_A further into $\mathcal{D}((\#A)^\Omega)$ and a finite product, but for k-tuples with a prime size, this creates a pole and zero respectively at $s = 1$ in those factors. Still, considering $\mathcal{D}((\#A)^\Omega)$ contains no information on the structure of A outside of its size, we should anticipate the finite product in that factorization to possess some structural information akin to M_A. As M_A and N_A are analogs for w_A and u_A, we compute their values at 1 for some admissible, non-residually-equivalent k-tuples, where the p column gives those primes over which $w_A(p) \neq 1$. Computations for M_A were made entirely in the rationals before rounding and so should be accurate. Computations for N_A were made using the product over the first 10000 primes before rounding, and so there is a possibility of error. We also include the corresponding C_As for reference, those being the reciprocals of the N_As. TODO: is there a dataset of these constants available somewhere? If not we could go high precision and provide it.

A	p	$M_A(1)$	$N_A(1)$	C_A
$\{0,2\}$	2	0.66666667	0.75738949	1.32032464
$\{0,10\}$	2,5	0.59259259	0.56804211	1.76043285
$\{0,6\}$	2,3	0.53333333	0.37869474	2.64064928
$\{0,2,6,8\}$	2,3	0.45714286	0.24089421	4.15119988
$\{0,8,12,30\}$	2,3,5,11	0.41975814	0.10539121	9.48845686
$\{0,6,12,18\}$	2,3	0.41558441	0.12044710	8.30239975
$\{0,2,6,8,30\}$	2,3,5,7,11	0.38614839	0.05639911	17.73077650
$\{0,18,30,60\}$	2,3,5,7	0.35463203	0.03011178	33.20959900
$\{0,60,120,180\}$	2,3,5	0.34996582	0.03011178	33.20959900

The table is sorted by $M_A(1)$, and at first glance it appears that $N_A(1)$ follows in step. The entries $\{0, 8, 12, 30\}$ and $\{0, 6, 12, 18\}$ have been added to dispel this. TODO: I don't think this next thing is really worth saying if we don't have anything else to say about it. Indeed, we have for k-tuples A and B with $\#A = \#B = c$,

$$\begin{aligned}
&M_A(1) < M_B(1) \\
\iff\ & 1 < \prod_p \frac{1 - w_A(p)p^{-1}}{1 - w_B(p)p^{-1}} \\
\iff\ & 0 < \sum_p \left[\log(1 - w_A(p)p^{-1}) - \log(1 - w_B(p)p^{-1})\right] \\
\iff\ & 0 < \sum_p \sum_{k \geq 1} \frac{w_B(p)^k - w_A(p)^k}{kp^k} \\
\iff\ & 0 < \sum_{k \geq 1} \frac{1}{k} \sum_p \frac{w_B(p)^k - w_A(p)^k}{p^k}.
\end{aligned}$$

Following the same proceedure for $N_A(1)$ gives

$$\begin{aligned}
&N_A(1) < N_B(1) \\
\iff\ & 0 < \sum_{k \geq 1} \frac{c^k}{k} \sum_p \frac{w_B(p)^k - w_A(p)^k}{p^k},
\end{aligned}$$

so the ordering of Ns is essentially that of the Ms, but with a higher weight given to higher powers of each prime.

We see also that the final two entries for $N_A(1)$ in the table appear to be exactly equal. It is not a rounding error: they are identical. This, along with their corresponding k-tuples having the same size, means that the Hardy-Littlewood Conjecture predicts these k-tuples to have identical limiting frequency, despite their residual inequivalence.

Just as the natural density of w_A can be expressed in terms of $M_A(1)$, we may express the limiting behavior of the summatory function of u_A via $N_A(1)$.

Theorem 2.13. *For any admissible k-tuple A, we have*

$$\frac{1}{x \log^{\#A-1} x} \sum_{n \leq x} u_A(n) \sim \frac{N_A(1)}{(\#A - 1)!}.$$

Proof. This follows immediately from theorem 5.2 of [4]. □

3 Dump zone

3.1 Unplaced results

Theorem 3.1. *For any admissible k-tuple A, we have*

$$u_A(n) \leq \varphi(n).$$

Proof. Let $A \in \mathbb{K}$, and let n be a natural number with factorization

$$n = \prod_{p_i | n} p_i^{a_i}.$$

Then

$$\begin{aligned} u_A(n) &= u_A\left(\prod_{p_i|n} p_i^{a_i}\right) \\ &= \prod_{p_i|n} u_A(p_i)^{a_i} \\ &\leq \prod_{p_i|n} (p_i - 1)^{a_i} \\ &\leq \prod_{p_i|n} (p_i - 1) \cdot p_i^{a_i - i} \\ &= \varphi(n). \end{aligned}$$

□

Definition 3.2.

$$\Psi_A(x) = \sum_{n \leq x} \Lambda(n) w_A(n).$$

Theorem 3.3. *For any admissible k-tuple A, we have*

$$\Psi_A(x) = \Psi(x) - \#S_A \log x + O(1).$$

Proof. We have

$$\Psi(x) - \Psi_A(x) = \sum_{n \leq x} \Lambda(n)(1 - w_A(n)).$$

Now, for composite n we have $\Lambda(n) = 0$, and for $n = p^k$ where $p \notin S_A$, we have $1 - w_A(n) = 0$, so the summand only takes on values at powers of $p \in S_A$, and we have

$$\begin{aligned}
\Psi(x) - \Psi_A(x) &= \sum_{p \in S_A} \log(p) \sum_{k \le \log_p x} (1 - w_A(p)^k) \\
&= \sum_{p \in S_A} \log(p) \sum_{k \le \log_p x} 1 + O\left(\sum_{p \in S_A} \log(p) \sum_{k \le \log_p x} w_A(p)^k \right) \\
&= \sum_{p \in S_A} \log(p) \sum_{k \le \log_p x} 1 + O(1) \\
&= \sum_{p \in S_A} \log(p)(\log_p(x) + O(1)) + O(1) \\
&= \sum_{p \in S_A} \log(x) + O(1) \\
&= \#S_A \log(x) + O(1).
\end{aligned}$$

□

Silly proof. We apply Perron's formula and obtain the following where the stars indicate the summation should be multiplied by 1/2 when $x \in \mathbb{Z}$.

$$\begin{aligned}
\Psi_A^*(x) &= \sum_{n \le x}^{*} \Lambda(n) w_A(n) \\
&= \frac{1}{2\pi i} \int_{c - i\infty}^{c + i\infty} \mathcal{D}(\Lambda \cdot w_A)(z) \frac{x^z}{z} dz,
\end{aligned}$$

where $c > 1$. Because w_A is completely multiplicative, we have

$$\begin{aligned}
\mathcal{D}(\Lambda \cdot w_A)(z) &= -\frac{d}{dz} \log(\mathcal{D}(w_A)(z)) \\
&= -\frac{d}{dz} \log(\zeta(z) M_A(z)) \\
&= -\frac{d}{dz} \log(\zeta(z)) - \frac{d}{dz} \log(M_A(z)) \\
&= \mathcal{D}(\Lambda)(z) - \frac{d}{dz} \log(M_A(z))
\end{aligned}$$

Substituting this into our integral, we then have

$$\begin{aligned}\Psi_A^*(x) &= \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\left[\mathcal{D}(\Lambda)(z) - \frac{d}{dz}\log M_A(z)\right]\frac{x^z}{z}dz \\ &= \Psi^*(x) - \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\left[\frac{d}{dz}\log M_A(z)\right]\frac{x^z}{z}dz.\end{aligned}$$

The derivative can be expanded as

$$\begin{aligned}\frac{d}{dz}\log M_A(z) &= \frac{d}{dz}\sum_{p\in S_A}\left[\log(1-p^{-z}) - \log(1-w_A(p)p^{-z})\right] \\ &= \sum_{p\in S_A}\left[\frac{\log(p)p^{-z}}{1-p^{-z}} - \frac{w_A(p)\log(p)p^{-z}}{1-w_A(p)p^{-z}}\right] \\ &= \sum_{p\in S_A}\left[\log(p)\sum_{k\geq 1}p^{-kz}(1-w_A^k(p))\right].\end{aligned}$$

(TODO: Is this a stupid way to eval this? There's probably a more direct route, but integrating series makes everything easy except for convergence.)

Now, substituting this into our above integral and distributing the integral over all sums (TODO convergence), we have

$$\Psi_A^*(x) = \Psi^*(x) - \frac{1}{2\pi i}\sum_{p\in S_A}\log p\sum_{k\geq 1}\int_{c-i\infty}^{c+i\infty}\frac{(x/p^k)^z}{z}dz.$$

When $\log_p(x) < k$ we have $x/p^k < 1$, and we may consider the integral to be closed along a right semicircle (just trust me dawg, but actually this is a critical detail). The integrand is analytic within that half-plane, so the integrals are all 0.

When $\log_p(x) > k$, we may then consider the integral to be along a left closed semicircle, so the integral evaluates to $2\pi i \cdot \mathrm{Res}((x/p^k)^z/z, 0) = 2\pi i$.

When $\log_p(x) = k$, the integral may be directly evaluated on the path $z = c + it$ as the vertical bound go symmetrically to infinity, and yields πi.

Accounting for each case, we have

$$\begin{aligned}\Psi_A^*(x) &= \Psi^*(x) - \sum_{p\in S_A} \log p \sideset{}{^*}\sum_{k\le \log_p x} 1 \\ &= \Psi^*(x) - \sum_{p\in S_A} \log p(\log_p(x) + O(1)) \\ &= \Psi^*(x) - \#S_A \log x + O(1).\end{aligned}$$

Now, the change in order $d(x)$ created by removing the stars when $x \in \mathbb{Z}$ is

$$d(x) = (\Psi(x) - \Psi_A(x)) - (\Psi^*(x) - \Psi_A^*(x)) = \frac{1}{2}\Lambda(x)(1 - w_A(x)).$$

If x is a power of a prime $p \in S_A$, d is bounded by the max of the logs of those finitely many primes. If x is a power of any other prime, we have $w_A(x) = 1$, so $d = 0$. If x is any other number, $\Lambda(x) = 0$, so $d = 0$. Then $d(x) = O(1)$, and the result follows. □

Combining this with the Prime Number Theorem in the form $\Psi(x) \sim x$ gives $\Psi_A(x) \sim x$.

3.2 Notes/TODO

1. The fundamental result we require is a relationship, asymptotic or otherwise, between u or w and the standard Ψ_A. Such a result would likely lead to proof of HLC.

2. We have $u_A * \phi_A = N = \phi_A^0$. That is, u_A jumps between levels of phi. What functions jump between other levels?

3. Formula for partial quotas in terms of full quotas:

$$\varphi_A^q = \sum_{k=q}^{\#A} (-1)^{k-q} \binom{k-1}{k-q} \sum_{\substack{B\subset A \\ \#B=k}} \varphi_B$$

   I think we should be getting a collapse down to $N = N$ for $q = 0$. Write a proof of this. This seems quite powerful. Can we find the order of 1 quotas in a simple way? Can this be inverted to express higher quotas in terms of lower? Does this allow expressing units in terms of one another?

4. Try looking into $\sum_{n \leq x} \Lambda(n)^{u_A(n)}$. For unmodulated primes this gives the correct number of lambdas, For modulated I think the power is reduced based on the chance that some member of the product for the standard Psi is zero. Idk, this is pretty vague, but it has a nice feeling

5. Maybe the relationship between the standard Ψ_A definition and $\zeta^{\#A}$ holds between a u/w modified version of Ψ and ζ? My instinct is that w_A represents some kind of order reduction.

   We know the w modified Ψ, say $\overline{\Psi}_A$ has the same limiting behavior as Ψ itself, so it's not going to reduce the order, but maybe showing $\Psi_A(x) \sim C_A \overline{\Psi}_A(x)$ will be easier?

6. Is there a meaning that can be attributed to simple operations on modulator constants? I think the only way to get milage out of this is to understand the probibalistic perspective from whence HLC came.

7. Continue to hunt for u and w series relations

8. Compute the ratio of the $0, 2$ and $0, 6$ HLC constants. Answer: $1/2$ (just requires cancellations) For equal tuple size it's rational.

9. Double check Granville actually implies the thing. Oh, just noticed he is the pretentious theory guy.

10. We should check if any results require finiteness on k-tuple

11. φ_A appears to be equivalent to a discrete convolution. Is that worth thinking about? Can we use that to improve the main proofs? What else can be done with this?

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